

Program: Diploma in Microelectronics	
Course Code: 5493B	Course Title: Digital CMOS VLSI
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To learn the fundamentals of VLSI design
- To familiarize the implementation of static and dynamic CMOS logic design.
- To gain knowledge in arithmetic building blocks and memory architectures.

Course Prerequisites:

Topic	Course code	Course Title	Semester
IC fabrication design rules		Introduction to IC Fabrication	3
Basics of VLSI design		VLSI Design	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the various methodologies in ASIC and FPGA design.	16	Understanding
CO2	Design VLSI Logic circuits with various MOSFET logic families.	12	Applying
CO3	Compare different types of memory elements.	14	Understanding
CO4	Design and analyse data path elements such as Adders and multipliers.	16	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the various methodologies in ASIC and FPGA design.		
M1.01	Differentiate between different kinds of ASICs	3	Understanding
M1.02	Explain the processes involved in SoC design.	3	Understanding
M1.03	Outline the FPGA design flow	4	Understanding
M1.04	Compare the speed, power and area considerations in VLSI	6	Understanding
Contents: VLSI Design Methodologies. Introduction: Moore's law .ASIC design, Full custom ASICs, Standard cell based ASICs, Gate array based ASICs, SoCs, FPGA devices, FPGA Design flows, Top-Down and Bottom-Up design methodologies. Speed power and area considerations in VLSI design			
CO2	Design VLSI Logic circuits with various MOSFET logic families.		
M2.01	Explain static and transient analysis of CMOS inverter	3	Understanding
M2.02	Realize the given logic function using static CMOS logic and transmission gate logic.	3	Applying
M2.03	Compare and contrast static and dynamic CMOS logic.	2	Understanding
M2.04	Realize the given logic function using pass transistor logic and transmission gate logic.	4	Applying
	Series Test - I	1	

Contents:**Static CMOS Logic Design**

MOSFET Logic Design - NMOS Inverter (Static analysis only), basic logic gates, CMOS logic, Static and transient analysis of CMOS inverter. Realization of logic functions with static CMOS logic, Pass transistor logic, and transmission gate logic

CO3	Compare different types of memory elements.		
M3.01	Explain the working of dynamic logic design and its drawbacks.	3	Understanding
M3.02	Explain NP domino logic.	5	Understanding
M3.03	Realize OR, NOR and NAND ROM cells.	4	Applying
M3.04	Compare and contrast SRAM and DRAM	2	Understanding

Contents:**Dynamic logic Design and Storage Cells**

Dynamic Logic Design-Pre charge- Evaluate logic, Domino Logic, NP domino logic.
Read Only Memory-4x4 MOS ROM Cell Arrays(OR,NOR,NAND)
Random Access Memory –SRAM-Six transistor CMOS SRAM cell,
DRAM –Three transistor and One transistor Dynamic Memory Cell..

CO4	Design and analyse data path elements such as Adders and multipliers.		
M4.01	Compare and contrast static adder and carry bypass adder.	2	Understanding
M4.02	Explain Linear carry select adder working.	5	Understanding
M4.03	Explain the working of square root carry select adder.	5	Understanding
M4.04	Explain an array multiplier.	4	Understanding
	Series Test - II	1	

Contents:

Arithmetic circuits

Adders: Static adder, Carry-Bypass adder, Linear Carry- Select adder, Square- root carry-select adder. Multipliers: Array multiplier.

Text /Reference:

T/R	BookTitle/Author
T1	Michael John Sebastian Smith, Application Specific Integrated Circuits, Pearson Education,2001.
T2	Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis & Design, McGraw-Hill, Third Ed., 2003.
R3	Jan M. Rabaey, Digital Integrated Circuits- A Design Perspective, Prentice Hall, Second Edition, 2005.
R4	Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education.
R5	CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009.

Online Resources:

Sl. No	Website Link
1	https://onlinecourses.nptel.ac.in/noc24_ee29/preview
2	https://pages.hmc.edu/harris/cmosvlsi/4e/index.html
3	http://www.vlsi-expert.com/
4	https://semiwiki.com/